

Chapter 3: Properties of Natural Resources

Textbook Exercise Solutions & Answers



1. Fill in the blanks with the appropriate term.

(a) The capacity of air to hold moisture depends upon the of the air.

temperature

(b) Water does not have a but has definite and

shape, volume, mass

(c) While freezing, the of water is lowered.

density

(d) soil has pH 7.

neutral

2. Why is it said that –

(a) Air is a homogeneous mixture of various gases.

1. Air is made of oxygen, nitrogen, carbondioxide, water vapour and dust particles.
2. All these are in different proportions. Therefore air is a mixture.

(b) Water is a universal solvent.

1. Water dissolves more substances than any other liquid. It dissolves solids, other liquids and even gases in it. Therefore, it is called as Universal solvent.
2. Water is used as a solvent in factories, laboratories, foodstuffs and in various types of biological processes occurring in the body such as digestion, excretion, etc.

(c) There is no alternative to water for cleaning purposes.

Water is a universal solvent. It dissolves more substances than any other liquid so there is no alternative to water for cleaning purposes.

3. What will happen if.....

(a) The amount of water vapour in the air increases.

We feel the dampness or the humidity.

(b) Only one crop is grown repeatedly in the soil.

The fertility of the soil decreases.

4. With whom should I pair up?

Column 'A' Column 'B' 1. Air a. Excretion 2. Water b. Scattering of light 3. Soil c. Plasticity

Column 'A'	Column 'B'
1. Air	b. Scattering of light
2. Water	a. Excretion
3. Soil	c. Plasticity

5. State whether the following statements are true or false.

(a) Sandy soil has low capacity for holding water.

True

(b) Sea water is a bad conductor of electricity.

False. Sea water is a good conductor of electricity

(c) The substance in which a solute dissolves is called a solvent.

True

(d) The pressure exerted by air is called atmospheric pressure.

True

6. Explain the picture in your own words.

- Diagram (a): Water is filled in the crack.
- Diagram (b): Water in the crack turns to ice due to decrease in temperature. When it turns to ice the volume increases and it expands. Due to expansion, the size of crack widens more than in diagram (a).

7. Write answers to the following questions in your own words.

(a) How is light scattered by the air?

Air is a mixture of very fine particles of some gases, dust, smoke and moisture. When rays of light fall on these minute particles, the particles spread the light in all the directions. This natural phenomenon is called scattering of light.

(b) Explain the various properties of water.

Various properties of water are as follows:

1. It is a fluid substance because it can flow.
2. It does not have its own shape, it takes the shape of a container in which it is kept.
3. It has a definite volume.
4. It is a transparent liquid and it has no colour.
5. It is tasteless and odourless.
6. It is a universal solvent.

(c) Why is the density of seawater is more than that of rainwater?

Seawater has many minerals dissolved in it as compared to rainwater. So due to more dissolved salts, density of seawater is more than that of rainwater.

(d) What is the importance of good soil structure?

The fertility of soil depends upon soil structure. So importance of good soil structure is as follows: (a) Roots get sufficient supply of oxygen, (b) Water drainage is good therefore the roots of plants grow well.

(e) What are the various uses of soil?

The uses of soil are as follows:

Plant conservation: To help plants grow. Water conservation: Soil holds water. As a result, by means of bunds and lakes, we can get water for use throughout the year.

Plasticity: (a) Soil can be given any required shape. This property of soil is called plasticity, (b) Because of its plasticity, we can use it to make articles of a variety of shapes, (c) These articles can be baked to make them hard, (d) Water storage earthen pots, earthen lamps, idols, bricks etc. are articles made from soil.

(f) What is the need and importance of soil testing from the point of view of farmers?

1. The properties of the various ingredients of soil can be determined by soil testing.
2. During soil testing, the soil is examined for colour, texture and the proportion of organic matter in it.
3. Soil is tested to find out if there is a deficiency of any ingredients and to decide what measures should be taken to remove the deficiency. These information would enable the farmers to maintain the fertility of the soil.

(g) What is the importance of air in transmission of sound?

1. Air is useful as a medium for the transmission of sound. Density of the air changes with temperature.
2. If the density of air increases, we can hear the sound more loud and clear.

(h) Why should a glass bottle completely filled with water never be kept in a freezer?

1. Due to anomalous behaviour of water, when you cool water below 4°C , the volume of the water increases.
2. Therefore when the glass bottle completely filled with water is kept in a freezer, water expands on cooling and its volume increases so glass bottle breaks.
3. To avoid this, completely filled bottle is never kept in a freezer.